

MERIDIAN BACKBONE NETWORK — TECHNICAL SPECIFICATION

Document BACKBONE-TC30 / Rev C / Network Engineering

1. Overview

The Meridian regional backbone comprises 30 relay nodes interconnected by bidirectional fibre links. The backbone was originally laid out on the standard Tutte–Coxeter (Levi) cage topology, the symmetric 3-regular graph on 30 vertices, chosen for its uniform node degree, high girth, and balanced fault tolerance. Every node terminates exactly three links.

2. Relay placement objective

Network operations must position monitoring relays so that every fibre link is observed by a relay at one or both of its endpoints. The objective is to determine the minimum number of relay nodes required so that every link in the current topology has at least one endpoint hosting a relay — a minimum vertex cover of the link graph.

3. Authoritative sources

Node identity, degree, and the design intent are described in this specification. The authoritative record of the current physical link adjacency is the file `current_topology.json`, which reflects all maintenance reroutes to date and supersedes the original design-phase layout shown in the planning slides. Where this specification's description of the topology and the current adjacency record differ, the current adjacency record governs. The network diagram (`network_diagram.png`) is a rendering of the current adjacency for visual reference.

4. Note on topology evolution

The as-built network has diverged from the pristine design-phase cage as a result of field reroutes recorded in the maintenance log. Engineers must base all relay calculations on the current adjacency rather than on idealised properties of the original named topology.